Supplementary Materials

Transfer learning from computed stability data for nanobody melting-temperature prediction

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Supplementary methods

Additional details for the heterogeneous MD data set

The heterogeneous data set was assembled from a SAbDab summary and the corresponding structures [1]. For each PDB identifier, we selected the first heavy-chain entry whose chain was present in the structure file. This rule yielded 1143 samples with a usable sequence, topology, trajectory, and native-contact label, representing 833 unique amino-acid sequences. Ninety-seven sequences occurred in more than one PDB entry and accounted for 407 samples. We calculated native-contact values separately for each structure and sampled them individually during training; duplicate sequences were not collapsed, and the model did not receive the PDB identifier. Eleven sequences exceeded the 158-residue model input and were truncated at the end.

Structures were cleaned and missing atoms were added with PDBFixer. The preparation first attempted protonation at pH 7.4 with pdb2pqr and PROPKA [2, 3]. If that step was unavailable or failed, the workflow used the geometry-based `pdb4amber --reduce` fallback, which does not apply the requested pH. If Amber conversion also failed, the PDBFixer output was retained. The available records do not show which route was used for each structure. The termini were left uncapped. Each protein was solvated in OPC water with 15 Å padding and 0.15 M NaCl and described with ff19SB [4, 5]. The 300-K branch comprised 1 ns NVT and 1 ns NPT with a 2-fs timestep. Two 1-ns restrained NVT stages at 400 K preceded unrestrained 400-K NVT production scheduled for 100 ns. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps^{-1} friction [6].

The contact calculation used all backbone heavy-atom pairs separated by more than three residues and within 0.45 nm in the first production frame, as for the mutation scans. Q was averaged over the final 30 ns of production, that is the last 300 frames of 100-ps output. The two MD data sets therefore used the same Q definition, contact set, native reference, and averaging window, and differed only in the variants they covered. Supplementary Fig. S1 shows the relation between sequence length and raw Q.

FEP variant and charge-change checks

The FEP tables used here comprised Ala, Asp, Gln, and Ile scans. Raw Phe scans existed but were not included in the processed tables, and the original reason for their omission was not recorded. Our conclusions therefore apply to the four-scan tables. For charge-changing mutations in a cubic 70-Å box and a solvent dielectric constant of 78, we tested an analytical periodic net-charge correction of $0.086(\Delta q)^2$ kcal mol^{-1}. We applied the correction before repeating the sign reversal, robust scaling, Yeo–Johnson transform with standardization, and min–max scaling used for the training labels (Supplementary Fig. S2b).

Model-size checks

We fine-tuned the 35M- and 650M-parameter ESM2 encoders without size-specific hyperparameter searches. For both sizes, the Tm-only model used the shared architecture, dropout 0.05, and an encoder learning rate of 1×10^{-4}. The FEP model used the shared architecture with separate 1MEL and 4IDL heads, $n = 320$ samples per structure, dropout 0.15, and an encoder learning rate of 3×10^{-5}. Both conditions used a non-encoder learning rate of 3×10^{-4}, weight decay 0.04, and ten training seeds.

References

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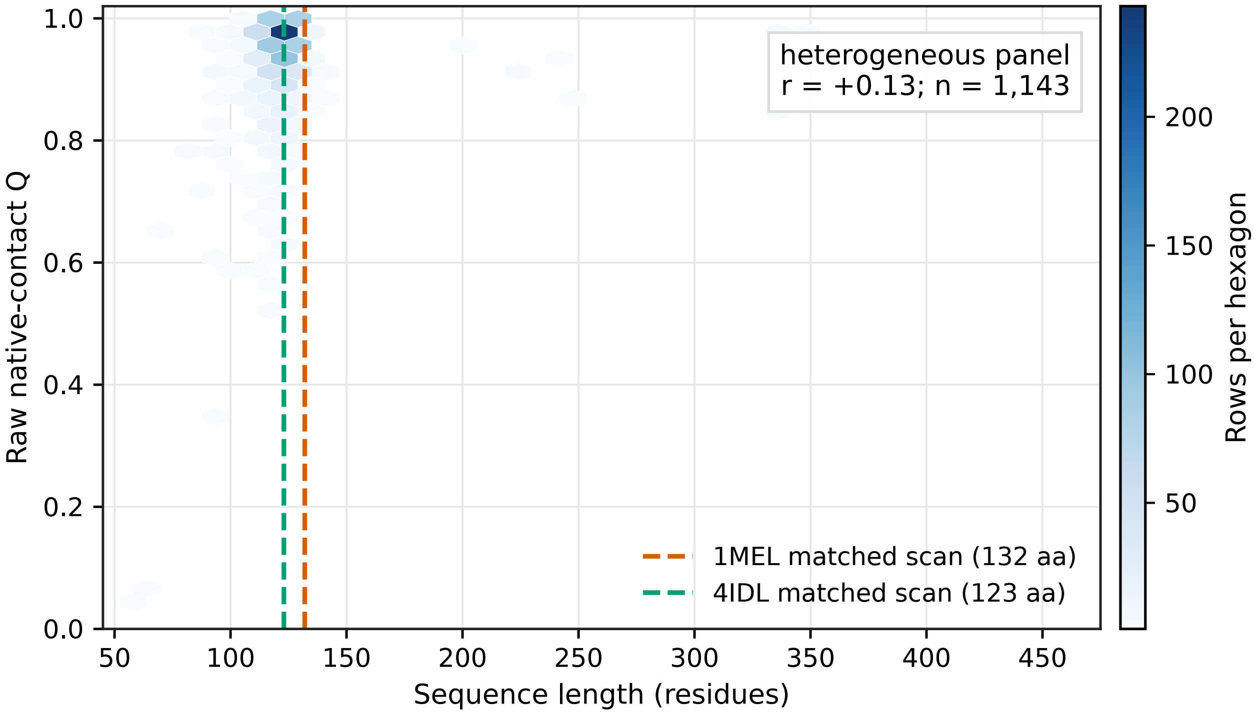

Figure S1 Sequence length and native-contact Q in the heterogeneous MD data set. Hexagons show sample density, and dashed lines mark the fixed sequence lengths of the 1MEL and 4IDL mutation scans. The association is descriptive and does not show that length caused the difference between the two MD data sets.

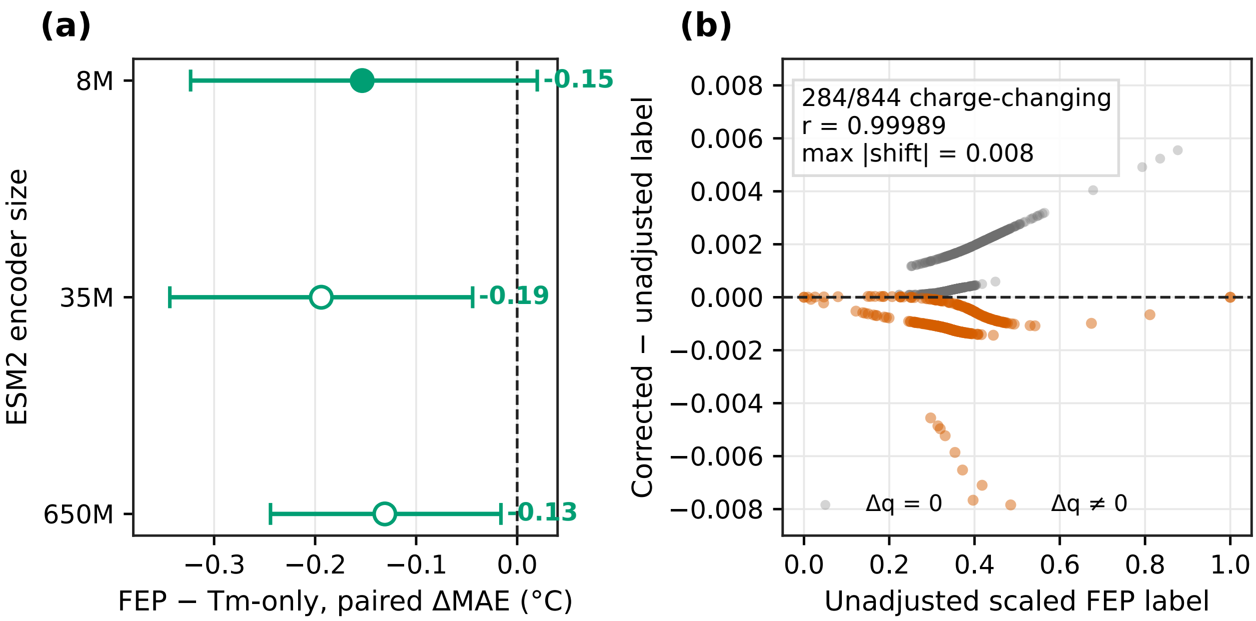

Figure S2 Additional checks of the FEP result. **(a)** Paired FEP-minus-Tm-only MAE for fine-tuned ESM2 encoders. Negative values mean lower Tm error with FEP. The 8M point is the five-seed model selected by the two-stage search. The 35M and 650M points are ten-seed checks without size-specific tuning. Whiskers are 95% paired bootstrap intervals over the 396 Tm test sequences. **(b)** Change in each scaled FEP label after the analytical periodic net-charge correction at dielectric 78.

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